TiN Oxidation Comparison

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September 27, 2019

Abstract

This memo is to cover two results: First, we compare a TiN device when it had just been fabricated, and when it had been sitting in the air for a year and a half. The film deposited was 100 cycles at 270C. Second, we compare a 50x 270C DC sample that had been left in the air for around 6 months with its behavior just out of fabrication and one month after fabrication.

1 DC Comparison

We compare one chip with itself as a function of how much time it has spent sitting in air. The chips are close to each other, albeit with the 6-month chip having a slightly lower T_c and a slightly higher R_n.

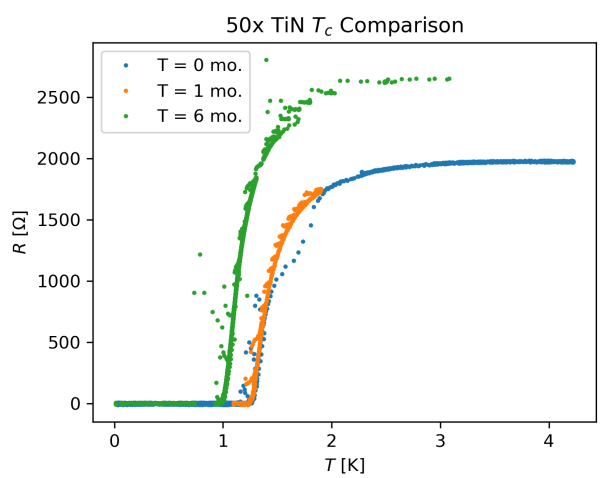

Figure 1: Comparison of DC T_c plots for 50x DC chip

2 RF Comparison

In the second run of testing, fewer resonators were found than the first run of testing, although the toneslist in the first test made clear a difference between deep resonators and shallow resonators, so it is possible we only found the deep resonators the second time. This means that the kid numbers are not necessarily the same, despite us matching resonant frequencies

2.1 RF T_c

We first compare the df/f sweeps for both chips, and see that they are quite similar.

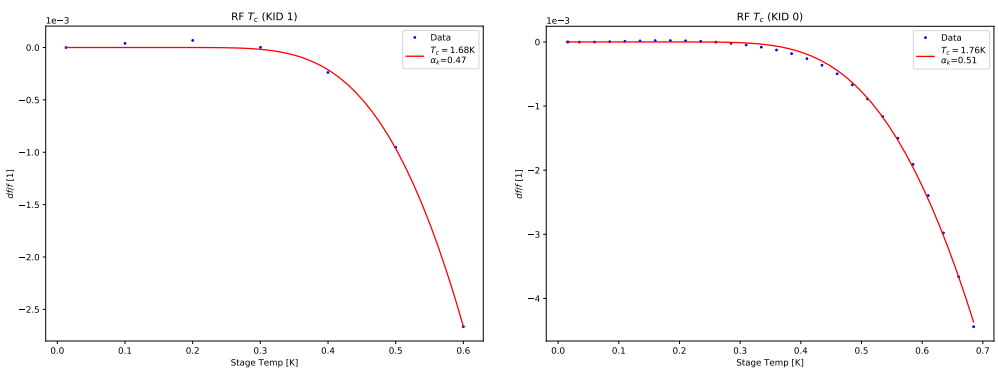

Figure 2: Comparison of RF T_c plots. (Left) old. (Right) new. Values were comparable between the two runs.

2.2 Q_i Values

Next, we compare the quality factors between the two chips, and see that they are similar. Note that we do not see as many resonators in the second test, so the overall number will be different.

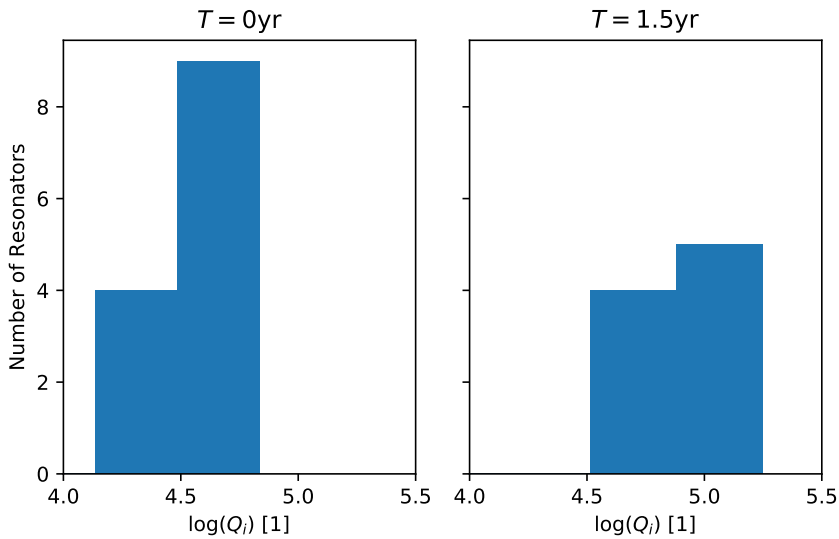

Figure 3: Q_i at $T_{\text{stage}} = 100$mK.

2.3 Noise

In terms of the noise analysis, we see things largely stay constant, although we took the new data without the pulse tube on, due to large pickup.

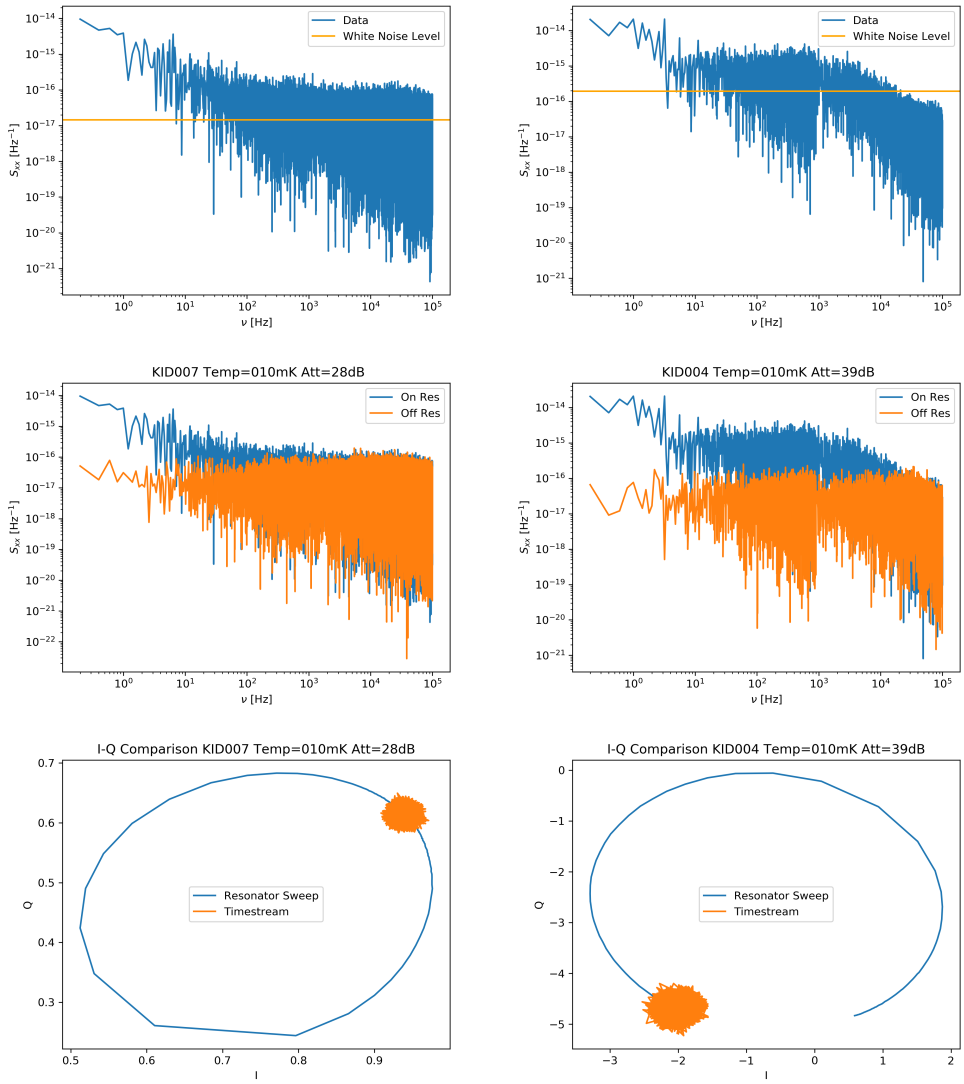

Figure 4: Comparison of noise behavior. (Left) old. (Right) new.